

Nitanshu Tak

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EDUCATION

University of Petroleum and Energy Studies

BTech. CSE, Cloud Computing and Virtualization Technology

Aug. 2023 – May 2027

CGPA: 8.02

EXPERIENCE

Suratec Co., Ltd.

Backend & Cloud Intern

May 2026 – Present

Thailand (Remote)

- Engineered a millisecond-accurate sensor-video synchronization engine using binary search ($O(\log n)$) to align BLE telemetry with video playback, validated across 4 real-device datasets at 27–67Hz with avg error of **5.7ms** against a 50ms tolerance threshold.
- Designed and implemented a **RESTful API integration layer** connecting a React Native mobile app to a Laravel backend, covering authentication, multipart video upload, sensor ingestion, and metadata-linked phase detection across **5 live endpoints**.
- Authored a formal synchronization audit identifying **10 timing-drift defects** across the recording and playback pipeline; implemented and unit-tested a complete remediation layer with **230 automated tests at 100% pass rate**.
- Diagnosed and resolved critical production bugs including a Laravel datetime-overflow corrupting all sensor writes and a PHP-FPM misconfiguration silently blocking all video uploads, traced via SSH to exact controller-level root causes.

SapMen C.

SDE - Backend and DevOps Specialist

Mar 2026 – Present

Hybrid

- Built 20+ Next.js server actions in TypeScript and Prisma/MongoDB to drive a 7-state project request workflow (quotation, acceptance, contract, invoice) with server-side status transition guards.
- Developed a 3-step request creation flow with live ORG-ID verification and a service catalog fetched from a REST endpoint, plus a Cloudflare R2 upload pipeline for PDF and image attachments up to 20MB.
- Implemented a polling-based chat feature (3s interval) between consumer and provider roles, backed by a MongoDB-stored message thread scoped per request.
- Fixed a production routing bug in Next.js middleware causing cross-role session redirects, replacing full page reloads with `router.refresh()` to preserve client-side navigation state.

PearlThoughts

DevOps Engineer Intern

Feb 2026

Remote

- Engineered a centralized automation suite to bridge the gap between local development and cloud-scale deployments using Terraform for reproducible Infrastructure as Code (IaC) provisioning.
- Developed a robust library of Bash-driven system utilities that automate critical Linux administration, user management, and resource monitoring tasks to eliminate manual overhead.
- Streamlined application portability by architecting optimized Docker configurations, ensuring seamless environment consistency from local machines to the cloud.
- Integrated automated GitHub Actions workflows to enforce code quality through continuous integration, standardizing the deployment lifecycle for scalable cloud-native applications.

PingTV

Backend Engineering Intern (Project-Based)

Jan 2026

Remote

- Developed an **automated backend content moderation system** using **Firebase Cloud Functions** and Firestore triggers for real-time evaluation of user-generated content.
- Implemented **database-level enforcement controls** (`isPublished`, `moderationStatus`, `isFlagged`) to automatically block policy-violating content from publication and monetization.
- Built a **moderation audit logging pipeline** with post IDs, decision reasons, and server timestamps to ensure traceability and compliance readiness.
- Delivered production backend features via GitHub PRs, aligning platform safety infrastructure with **Stripe compliance requirements**.

- Engineered core dashboard modules (Sidebar, Inventory, Notifications) using **React, Next.js, and Tailwind CSS**, streamlining daily operations and improving platform stability.
- Architected scalable routing and a library of reusable components, reducing development overhead by **30%** and accelerating feature delivery across the application.
- Developed real-time analytics and automated inventory workflows, facilitating data-driven insights that improved executive decision-making speed by **45%**.
- Translated **Figma** prototypes into high-fidelity UIs and integrated backend APIs, optimizing frontend performance and reducing the design-to-production lifecycle by **30%**.

PROJECTS

SAL-SHIELD - AI & GIS Forest Infestation Early Warning Platform

Tech Stack: Python, Flask, TensorFlow, XGBoost, scikit-learn, React, Sentinel-2, QGIS | GitHub

- Designed and trained a dual ML pipeline - Random Forest (**88.8%**) and XGBoost (**97.5%**) on 21 Sentinel-2 satellite features, plus a custom MobileNetV2-based CNN (**92.7%**) on 691 field tree images - to classify infestation risk at both forest and tree scale.
- Invented the **Sal Borer Vulnerability Index (SBVI)**, an original composite geospatial risk score derived from NDVI, NDMI, NBR, and terrain factors across **4 temporal Sentinel-2 scenes** (Oct 2023 – Jun 2025), identifying **24 Very High Risk zones** in the Dehradun Forest Division.
- Architected a full-stack WebGIS platform (Flask REST API + React + Vite PWA) with Leaflet GIS map rendering, offline-capable field reporting via IndexedDB & Workbox, and real-time CNN inference completing tree health classification in **under 2 seconds**.
- Deployed zero-cost production infrastructure (Render + Vercel) targeting **HILL Conference 2026**; system works completely offline in zero-connectivity forest environments with automatic sync on signal recovery.

CloudDSL Compiler - Six-Phase DSL-to-Cloud Deployment Compiler

Tech Stack: Java 21, Maven, TypeScript, VS Code Extension API | GitHub | Patent Pending - UPES R&D

- Engineered a full six-phase compiler (Lexer → Parser → Semantic Analyser → AST → Optimizer → Code Generator) that transforms a declarative **.dsl** source file into **3 simultaneous production deployment artefacts**: Dockerfile, Kubernetes YAML (with HPA), and AWS ECS Fargate Bash script.
- Implemented a DFA-based tokenizer, EBNF recursive descent parser, and a semantic analyser enforcing **10 validation checks** with symbol table construction and aggregate error reporting.
- Built an optimisation pass for defaults injection and autoscale inference, reducing boilerplate by **>85%** - 8 lines of DSL replacing **>7KB** of hand-authored cloud config across 3 platforms.
- Shipped a **VS Code Language Extension** with TextMate syntax highlighting, code snippets, live diagnostics, and auto-compile on save; validated by a 50-case JUnit 5 test suite.

NitanshuOS - Custom Linux Distribution & Boot Pipeline

Tech Stack: Linux Kernel, Buildroot, GRUB, QEMU, Docker, AWS | GitHub

- Built a minimal Linux distribution from source, compiling a custom x86-64 kernel and generating a BusyBox-based root filesystem, reducing base system footprint by **>60%** compared to standard desktop distros.
- Automated a **5-stage bootable disk pipeline** (partitioning, filesystem creation, rootfs injection, GRUB setup, image packaging) enabling full system generation in **<10 minutes**.
- Containerized the OS build and validation workflow using Docker, achieving reproducible kernel builds and eliminating host-environment dependency issues.
- Validated end-to-end boot execution in QEMU and extended workflow to cloud by converting raw disk images to VMDK format for AWS AMI import experimentation.

TECHNICAL SKILLS

Languages: C, C++, Python, Java, JavaScript

Cloud & DevOps: AWS (EC2, S3, IAM, EKS, CloudWatch, Lambda), Docker, Kubernetes, Terraform, CI/CD Pipelines, Microservices, GitHub Actions

Backend & APIs: Flask, Laravel, REST APIs, JWT Authentication, Firebase Cloud Functions

AI/ML & Data: TensorFlow, Keras, scikit-learn, XGBoost, MobileNetV2, Isolation Forest, Pandas, QGIS, Rasterio

Security & IAM: Identity & Access Management (IAM), OAuth 2.0, OpenID Connect (OIDC), MFA

Frontend & PWA: React, Next.js, Vite, Tailwind CSS, Leaflet, Recharts, Workbox, IndexedDB

Compiler & Systems: Lexical Analysis (DFA), Recursive Descent Parsing (EBNF), AST Construction, Code Generation, OS Internals, Buildroot
Databases: MySQL, PostgreSQL, MongoDB, SQLite, Firestore
Tools: Git, GitHub, Linux CLI, VS Code, Postman, QEMU, Maven, Google Colab

CERTIFICATIONS

- **AWS Academy Certified Operations** - AWS Academy, Dec 2025 [Credential]
- **AWS Academy Certified Architecting** - AWS Academy, Nov 2025 [Credential]
- **OCI Certified DevOps Professional** - Oracle University, Oct 2025 [Credential]
- **OCI Certified Multicloud Architect Professional** - Oracle University, Sep 2025 [Credential]

ACHIEVEMENTS

- **Admitted - Algorverse AI Research Program, Summer 2026 Cohort** - Selected with scholarship offer for an international LLM & Applied ML Research cohort among applicants from multiple countries; declined due to existing professional and academic commitments.
- **Selected Contributor/Mentee - GirlScript Summer of Code 2026 (GSSoC'26)** - Accepted into both the **Open Source Track** and **AI/Agents Track**; contributing to impactful open-source projects and collaborative AI agent systems. [Profile]
- **University Rank 2 on Unstop** and placed in the **Global Top 0.21%** among participants; also nominated for **Unstop Talent Awards 2026**.
- **Google Cloud Arcade Facilitator Program 2025** - Achieved Diamond League and Trooper Tier with **14,852 points** through hands-on cloud architecture labs. [Post]
- **International Conference Presenter (ICATEET 2026)** - Presented research project "Helios: AI-Enabled Secure Cloud Infrastructure" with live system demonstration. [Project] — [Post]
- **Ideathon Finalist (Solo)** - "Kevlar" project shortlisted for final round; independently designed and presented solution to industry professionals. [Project] — [Post]
- **Hack For Green Bharat** - Ranked within **Top 100 teams** nationally among 7,000+ competing teams.
- **Open Source Contributor - Hacktoberfest** - Contributed to open-source repositories during Hacktoberfest; recognized for community-driven development contributions.
- **Campus Ambassador** - LetsUpgrade, Internshala Student Partner, IIT Roorkee E-Summit 2026, IIM Indore IRIS, IIT Delhi EDC.